

TASK 05-50-09-210-801-A

EFFECTIVITY: ALL

1. Overweight Landing - Inspection/Check

A. General

- (1) An overweight landing inspection is necessary when these two conditions occur:
 - (a) The aircraft lands with weight above the MLW;
 - (b) The crew cannot tell if the limit sink rate of 300 fpm was obeyed.
- (2) This procedure lets you use flight data source for the check of the loads at touchdown and compare it with the aircraft landing envelope. For this procedure, the vertical CG acceleration and the roll rate at touchdown are used.
- (3) These maintenance technicians are necessary to do this task:
 - 2 - Airframe (Structure)

B. References

REFERENCE	DESIGNATION
AMM MPP 32-11-01/400	MAIN-LANDING-GEAR SHOCK STRUT - REMOVAL/INSTALLATION
AMM MPP 32-11-13/400	MAIN-LANDING-GEAR LOCKING STAY - REMOVAL/INSTALLATION
AMM MPP 53-04-20/400	WING-TO-FUSELAGE FAIRING FORWARD PANEL I - REMOVAL/INSTALLATION
AMM MPP 53-04-30/400	WING-TO-FUSELAGE FAIRING CENTER PANEL - REMOVAL/INSTALLATION
AMM MPP 53-04-40/400	WING-TO-FUSELAGE FAIRING LH SIDE PANEL - REMOVAL/INSTALLATION
AMM MPP 57-55-09/400	OUTBOARD FLAP ROD ASSY - REMOVAL/INSTALLATION
AMM SDS 52-11-00/1	FORWARD PASSENGER DOOR
AMM SDS 52-31-00/1	FORWARD CARGO DOOR
AMM SDS 52-32-00/1	AFT CARGO DOOR
AMM SDS 52-41-00/1	FORWARD SERVICE DOOR
AMM SDS 52-42-00/1	AFT SERVICE DOOR
AMM TASK 05-50-03-200-801-A/600	Hard Landing and Off-Temperature Envelope - Inspection/Check
AMM TASK 07-10-01-500-801-A/200	Complete Aircraft Jacking - Maintenance Practices
AMM TASK 12-11-01-650-802-A/300	Fuel Tank - Pressure Defueling
AMM TASK 20-00-00-910-801-A/200	Aircraft Safety Procedures Before Maintenance - Standard Procedures
AMM TASK 20-00-00-910-802-A/200	Aircraft Safety Procedures After Maintenance - Standard Procedures
AMM TASK 31-31-00-970-806-A/200	DVDR System - Download Procedure with RDI Application

(Continued)

REFERENCE	DESIGNATION
AMM TASK 32-11-07-000-801-A/400	Main-Landing-Gear Wheel Axle - Removal
AMM TASK 32-11-07-400-801-A/400	Main-Landing-Gear Wheel Axle - Installation
AMM TASK 32-11-11-000-801-A/400	Main-Landing-Gear Side Stay - Removal
AMM TASK 32-35-00-710-801-A/500	Emergency-Extension System - Free Fall (On Jacks) - Operational Test
AMM TASK 32-49-11-000-801-A/400	Brake Assembly - Removal
AMM TASK 32-49-11-400-801-A/400	Brake Assembly - Installation
AMM TASK 32-61-07-820-801-A/200	Main-Landing-Gear Downlock Sensor (Clearance) - Adjustment
AMM TASK 32-62-03-820-801-A/200	Main-Landing-Gear Weight-Off-Wheels Sensor (Clearance) - Adjustment
AMM TASK 51-50-00-820-801-A/200	Aircraft Alignment Check
PW1900G PPMM PW1000G-A-72-00-00-09C-0B5A-A	Do An Inspection of the Engine After High Turbulence or Lateral Loads; Heavy or Overweight Landings

C. Overweight Landing

SUBTASK 210-002-A

- (1) The flowchart is a summary of the necessary actions when an overweight landing occurs. Refer to [Figure 601](#)
- (2) When the crew reports a landing with the weight above the MLW, and they cannot tell if the sink rate at touchdown was less than 300 fpm, do the Phase-I inspection immediately. If you do not find incorrect conditions during the Phase-I inspection, a deferrable period of 10 FC is permitted. Then, you can do one of these two procedures:

NOTE: No inspections are necessary for sink rates less than 300 fpm.

- (a) At the end of the deferrable period, do the Phase-II inspection. In this case, no flight data analysis is necessary. You must do Phase-III inspection only if you find an incorrect condition during the Phase-II inspection.
- (b) During the deferrable period, obey the instructions given in this section to analyze the flight data. Then, you must do the Phase-II inspection only if the data analysis shows that the loads exceeded the aircraft envelope for the MLG touchdown. The Phase-III inspection is applicable only if you find an incorrect condition during the Phase-II inspection.

NOTE: The flight data download must be done within a maximum of 20 FH. If not, the data may be missed due to the recording time of the unit.

- (3) The procedure to find an excess to the landing envelope when an overweight landing occurs and the related inspections are applicable only to the MLG. Do the hard landing inspection (AMM TASK 05-50-03-200-801-A/600) if the crew reports one of these two conditions:
 - The aircraft touches the ground with the NLG first.

- The crew reports a suspect hard landing in the NLG because of high pitch-angle variation.
- (4) Do a download of the DVDRs (AMM TASK 31-31-00-970-806-A/200):
- Vertical CG acceleration (NormAccel)
 - Air/Ground indication (Air/Gnd)
 - Gross weight (GrossWeight)
 - Roll (Roll)
 - Time (GMT)
 - Date (Day)
 - Flight number (FltNumber)
 - Pitch angle (Pitch)

NOTE: The data must be selected from the flight data source at a rate of 8 sps (samples per second).

- (5) The procedures given here let you find an envelope exceedance condition and put the related aircraft in a serviceable condition. If you will not do the Phase-III inspection, it is not necessary to send data to Embraer. But, if it is necessary to do the Phase-III inspection, send the data to Embraer for analysis (flight data source and a report of the damage found). The decision to put the aircraft back into service will then be made by Embraer together with you.
- (6) When the conditional inspection tells you to "examine" a component, you must look for the conditions that follow:
- Pulled-apart structure
 - Loose paint (paint flakes)
 - Discoloration
 - Distortion (twisted parts)
 - Wrinkles or buckles in the structure
 - Bent components
 - Fastener holes that become oversized
 - Loose fasteners
 - Fasteners that pulled out or are missing
 - Delamination
 - Fiber breakouts
 - Black powder residues around, and that trail from, fasteners in composites (indication of loose fasteners)
 - Misalignment

- Interference
- Nicks or gouges
- Scratches
- Other signs of damage

(7) All the inspections are classified as GVI.

(8) If during the inspection of the structures there are signs of damage related to the overweight landing event, contact Embraer for the airworthiness restoration of the related aircraft.

D. Overweight Landing Detection

SUBTASK 210-003-A

(1) The overweight detection for sink rates more than 300 fpm is the same of MLG roll rate analysis presented at hard landing task. Do (AMM TASK 05-50-03-200-801-A/600) SUBTASK 210-003-A.

E. Job Set-Up

SUBTASK 940-002-A

WARNING: MAKE SURE THAT THE AIRCRAFT IS IN A SAFE CONDITION BEFORE YOU DO THE MAINTENANCE PROCEDURES. THIS IS TO PREVENT INJURY TO PERSONS AND/OR DAMAGE TO THE EQUIPMENT.

(1) Do the procedure to make the aircraft safe for maintenance (AMM TASK 20-00-00-910-801-A/200).

SUBTASK 940-003-A

(2) Put all flight control surfaces in the neutral position.

F. Phase-I Inspection

SUBTASK 210-004-A

(1) Do a visual inspection of the items shown below. Do not disassemble them.

(2) Main Landing Gear

- (a) Examine the MLG wheels for distortion, bending, dents and fracture.
- (b) Examine the MLG tires for general condition and deflation.
- (c) Examine the brake assembly for damage, and the hoses for signs of fluid leakage.
- (d) Examine the MLG wheelwell for signs of fluid leaks.
- (e) Examine the MLG locking stay for distortion, bending and fracture.
- (f) Examine the upper and lower MLG side stays for distortion, bending, and fracture.
- (g) Examine the MLG shock absorber conditions.

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(h) Examine the upper and lower MLG trailing arm conditions.

(3) Fuselage

(a) Examine the external surface of the fuselage for fluid leakage.

(b) Examine the external areas near the wing stub for any signs of damage.

(c) Open and close the forward and aft passenger doors. Make sure that they operate correctly (AMM SDS 52-11-00/1 and AMM SDS 52-32-00/1).

(d) Open and close the forward and aft service doors. Make sure that they operate correctly (AMM SDS 52-41-00/1 and AMM SDS 52-42-00/1).

(e) Open and close the forward and aft cargo doors. Make sure that they operate correctly (AMM SDS 52-31-00/1 and AMM SDS 52-32-00/1).

(f) Externally examine the aft cargo door metallic and composite scuff-plate for signs of bend.

(g) Examine the tail cone and silencer.

(h) Examine the wing-to-fuselage fairing junction with the fuselage and look for any missing or damaged screw, missing sealant, and paint peel-off

(4) Wing

(a) Examine the MLG doors for misalignment and signs of damage.

(b) Examine the external surface of the wing for fuel or other fluid leakage.

NOTE: You must carefully examine the fuel-tank access-panel contours for fuel leakage.

(c) Examine the pylon fairings for signs damage.

(5) Flight Controls

(a) With the flaps, slats and spoilers fully retracted and ailerons, rudder and elevators in the neutral position, examine all surfaces for misalignment.

(b) With the flight controls fully deflected, examine the internal mechanisms for bent components.

(c) Examine all flight controls to make sure that they move freely.

NOTE: If necessary, use a flashlight to do the inspection of the inner side of the flap rod fairing for bent components.

(6) Engine

(a) Examine the inlet-cowl internal skin for signs of chafing against the fan blades.

G. Phase-II Inspection

SUBTASK 210-005-A

(1) Main Landing Gear

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- (a) Do an inspection of the lower skin of the rear fuselage around the jacking point before you put the aircraft on jacks. If there are signs of buckling, do not put the aircraft on jacks and speak to Embraer for more instructions.
- (b) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).
- (c) Do a check of the clearances of the proximity sensors. The sensor surfaces must be parallel.
 - 1 For the downlock sensor, refer to AMM TASK 32-61-07-820-801-A/200.
 - 2 For the weight-on-wheels sensor, refer to AMM TASK 32-62-03-820-801-A/200.
- (d) Remove the MLG side-stay lower bolt (AMM TASK 32-11-11-000-801-A/400) and examine for distortion, bending, and fracture.
- (e) Examine the side-stay lower bolt lug for distortion, bending, and fracture. Refer to [Figure 602, Sheet 1](#).
- (f) Examine the trunnion fitting. Refer to [Figure 602, Sheet 2](#).
- (g) Retract the landing gear and do the visual check of the retraction path for interferences, clearances, misalignments, malfunctions and unusual noise in the landing gear, landing gear bay and landing gear door mechanisms. Extend the landing gear and do the check for unusual noise, uplock and downlock mechanisms malfunction. Do the check for any unusual behavior.

NOTE: An irregular, non-smooth operation of the landing gear can show deformation of the shock absorber and rear pintle pin. They are the most critical parts when a hard landing event occurs. Do the check of their condition and correct alignment during MLG phase III inspection.
- (h) Do the retraction procedure again and extend the landing gear with the free-fall system. Do the check for any unusual behavior.
- (i) Remove the brake assembly (AMM TASK 32-49-11-000-801-A/400).
- (j) Examine the wheel axle to know if it is bent. Remove it, if necessary (AMM TASK 32-11-07-000-801-A/400).
- (k) If removed, install the wheel axle (AMM TASK 32-11-07-400-801-A/400).
- (l) Install the brake assembly (AMM TASK 32-49-11-400-801-A/400).
- (m) In case of tire burst during the event, do the check of the side-stay condition.

(2) Fuselage

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- (a) Externally examine the upper and lower fuselage-skin panels for loose or sheared rivets, structural damage, and signs of fuel and other fluid leakage.

NOTE: • For the inspection of the lower skin panels, you must remove the wing-to-fuselage fairings (AMM MPP 53-04-20/400, AMM MPP 53-04-30/400, and AMM MPP 53-04-40/400)

- If you find wrinkles in the skin panels, you must do an internal visual inspection on the fuselage.

- (b) Carefully examine all the systems inside the wing-to-fuselage fairing region and its fixation to the fuselage.
- (c) On the forward electronic compartment, examine the MAU/EICC and shelves for integrity.
- (d) On the middle electronic compartment, examine the electronic rack and shelves for integrity.
- (e) On the rear fuselage, examine the electronic racks and shelves for integrity.
- (f) Carefully examine the internal side of the fuselage in the attachment of the galley to the frame. Refer to [Figure 603, Sheet 1](#), [Figure 603, Sheet 2](#), and/or [Figure 603, Sheet 3](#).

NOTE: It is not necessary to remove the galley to get access to the inspection area.

(3) Wing

- (a) Examine wing spars II and III. Start from the MLG wheelwell. Remove the access panels and, with the inboard flap in the down position, also examine the lower skin adjacent to them.
- (b) Examine the upper and lower wing skin panels from the wing root to the wing tip.
- (c) Externally examine the wing, nacelles, and engine struts for loose or sheared rivets, structural damage, and signs of fuel and other fluid leakage.
- (d) Examine the wing-to-fuselage junction and the MLG wheelwell for damage. Look for flaked paint and loose or missing fasteners.

(4) Flight Controls

- (a) Do an inspection of the flap rod mechanism (AMM MPP 57-55-09/400).
- (b) With the flaps fully deflected, examine the slat mechanism for bent components.
- (c) With the flaps fully retracted, examine the flap and slat surfaces for any misalignment.
- (d) In the MLG compartment, do a check to see if the aileron cable and the pulleys are in the correct position.

(5) Engine

- (a) Do engine inspection as per (PW1900G PPMM PW1000G-A-72-00-00-09C-0B5A-A).

H. Phase-III Inspection

SUBTASK 210-006-A

- (1) Defuel the aircraft (AMM TASK 12-11-01-650-802-A/300).
- (2) Put the aircraft on jacks (AMM TASK 07-10-01-500-801-A/200).
- (3) Remove the MLG locking stay pins (AMM MPP 32-11-13/400) and examine for distortion, bending, and fracture.
- (4) Remove the MLG pintle pins (AMM MPP 32-11-01/400) and examine for distortion, bending, and fracture.
- (5) Examine the landing gear for correct operation of the free-fall system (AMM TASK 32-35-00-710-801-A/500).

NOTE: An irregular, not-smooth operation of the landing gear during free-fall can show deformation of the landing gear. You must then examine it for correct alignment.

- (6) Carefully examine the internal side of the CF II in the area below the floor cross beams and columns.
- (7) Do the aircraft alignment check (AMM TASK 51-50-00-820-801-A/200).

I. Job Close-Up

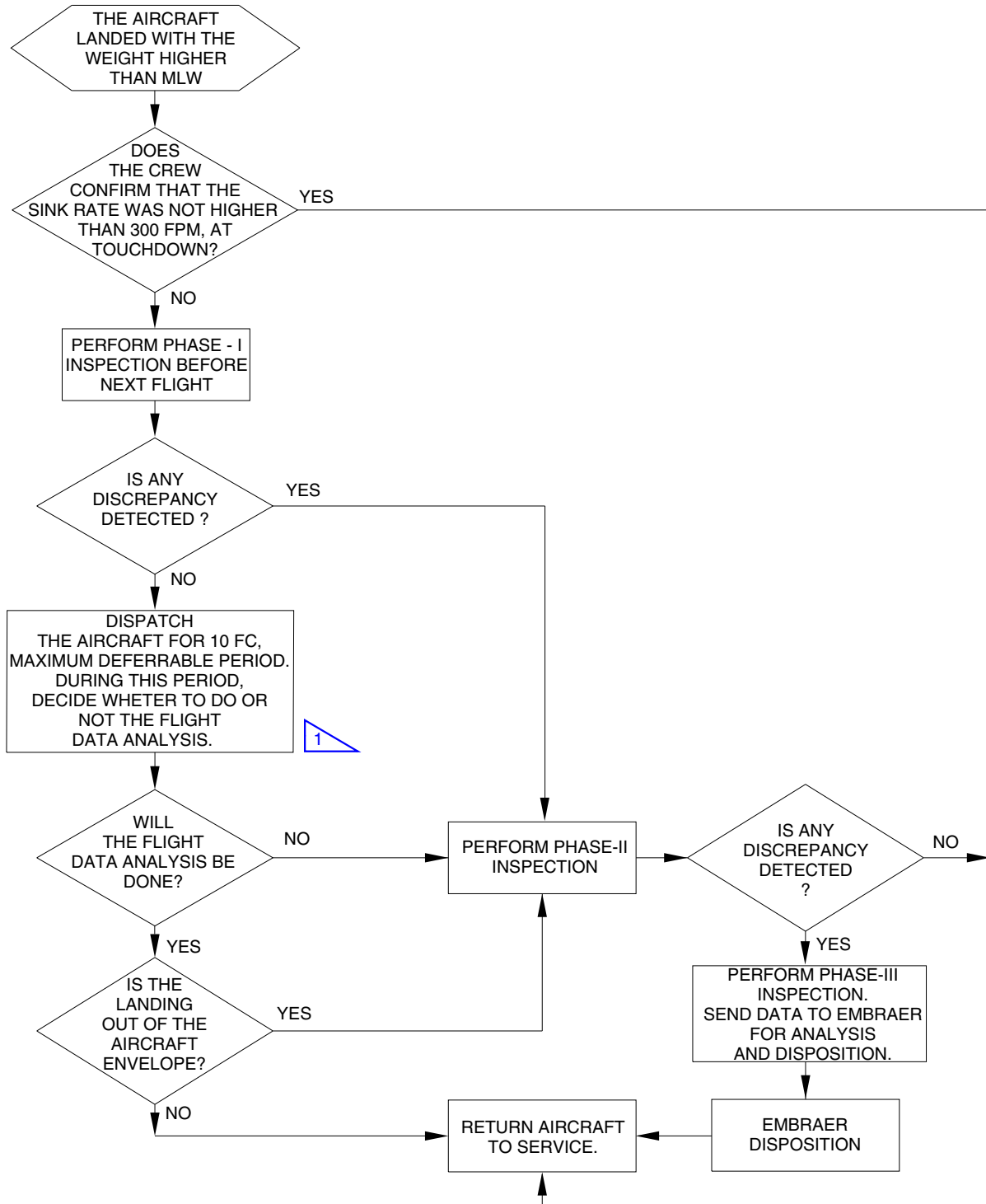
SUBTASK 840-002-A

- (1) Remove from the work area all tools, equipment, and materials that you used.
- (2) Do the procedure to restore the aircraft after the maintenance procedures (AMM TASK 20-00-00-910-802-A/200).

EFFECTIVITY: ALL

Overweight-Landing Inspection Flowchart

Figure 601



1 THE FLIGHT DATA DOWNLOAD SHALL BE PERFORMED IN A MAXIMUM OF 20FH. IF NOT, THE DATA CAN BE MISSED DUE TO THE RECORDING TIME OF THE UNIT.

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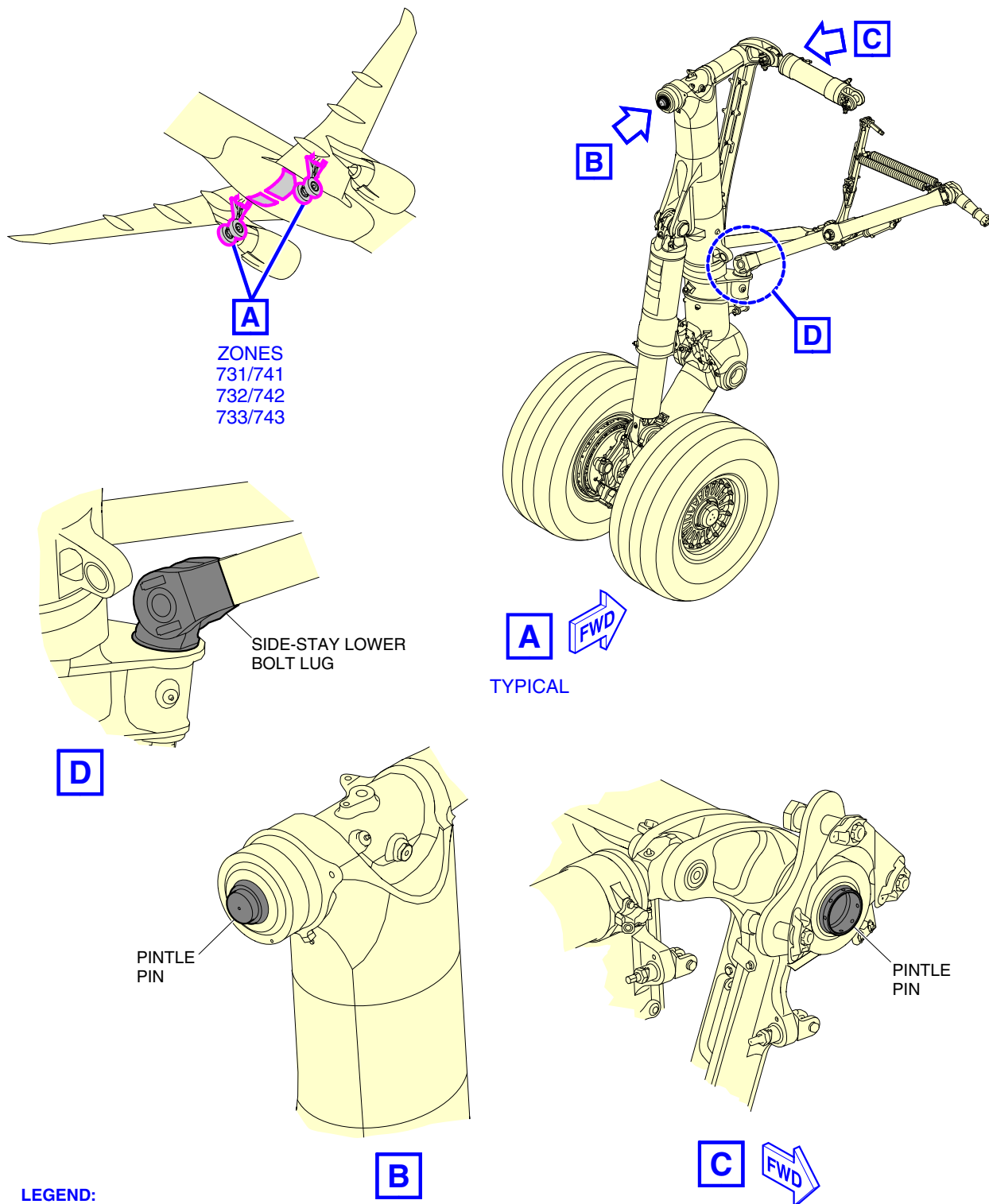
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EFFECTIVITY: ALL

MLG Identification and Structures

Figure 602 - Sheet 1



LEGEND:

■ INSPECTION AREA

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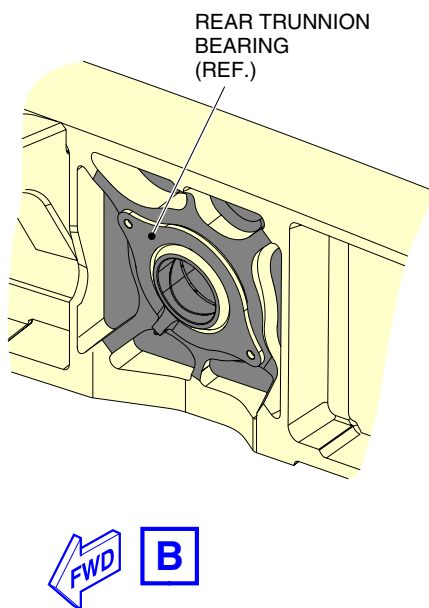
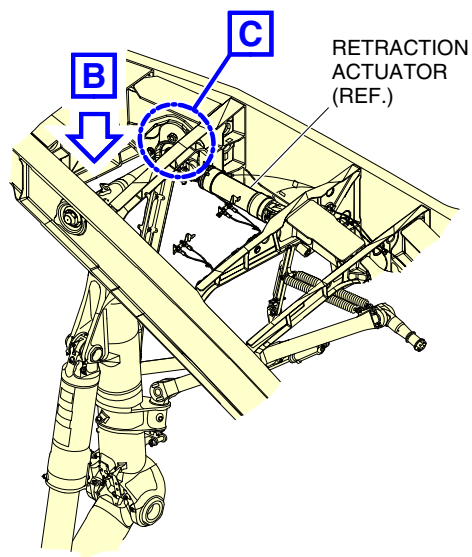
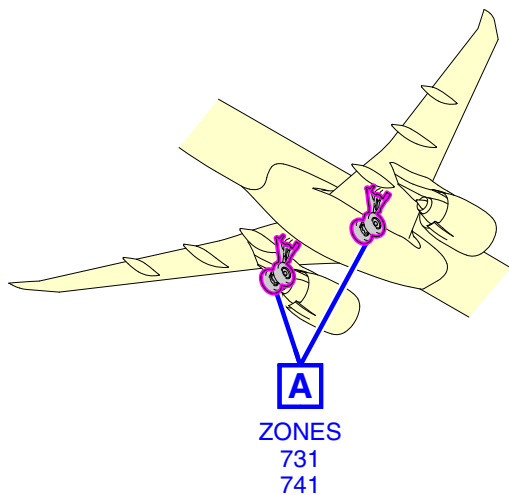
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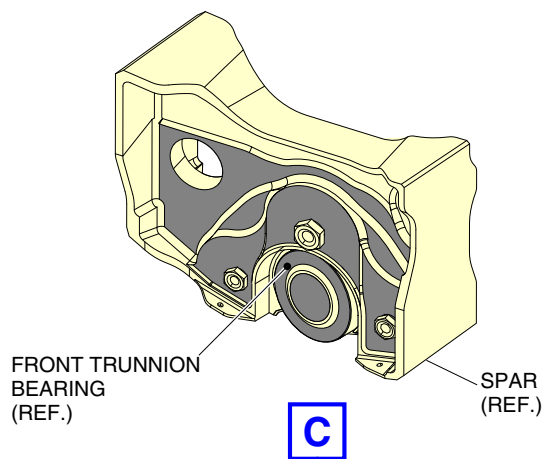
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MLG Identification and Structures

Figure 602 - Sheet 2



A
TYPICAL



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■ INSPECTION AREA

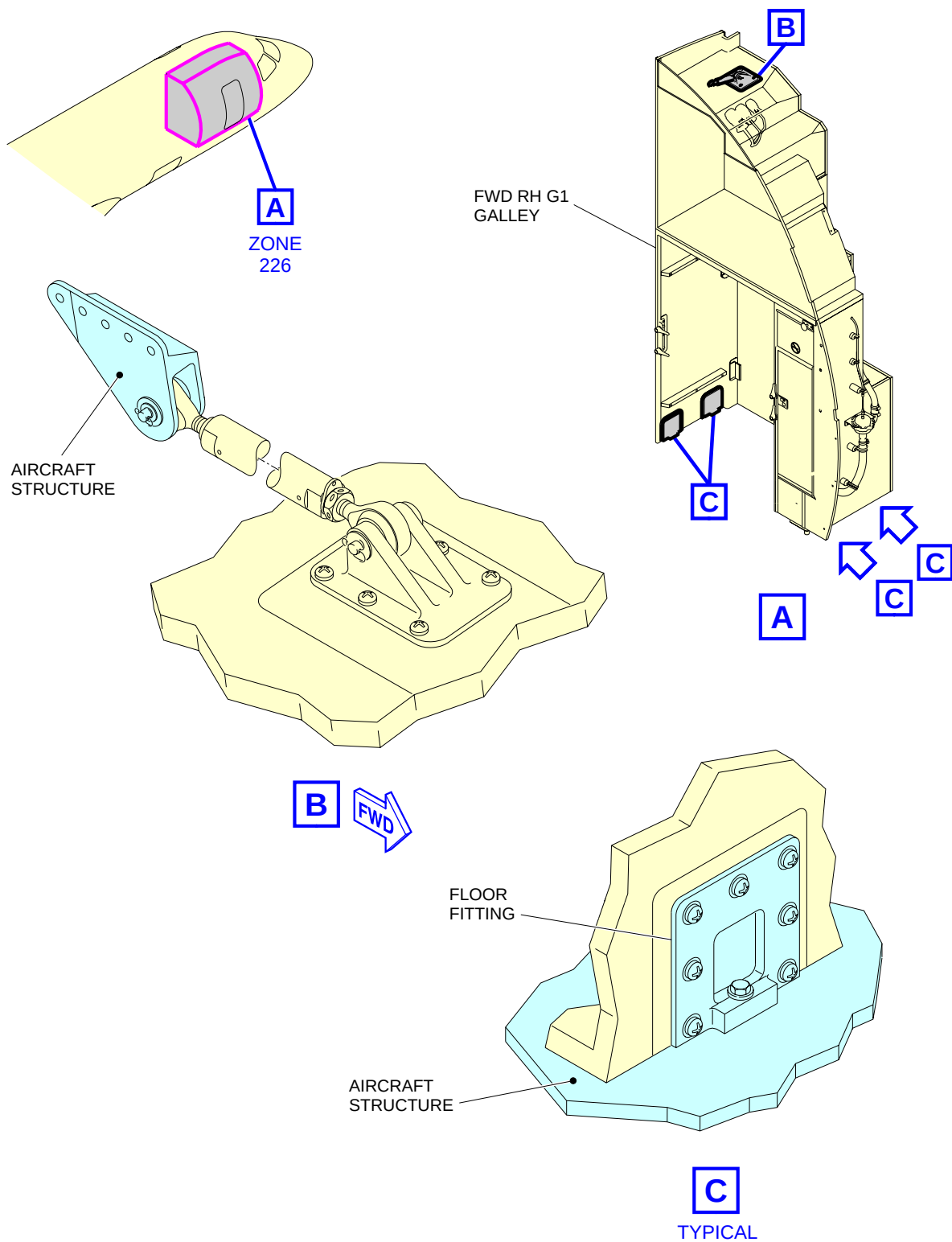
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EFFECTIVITY: ALL

Galley Attachments - Inspection

Figure 603 - Sheet 1



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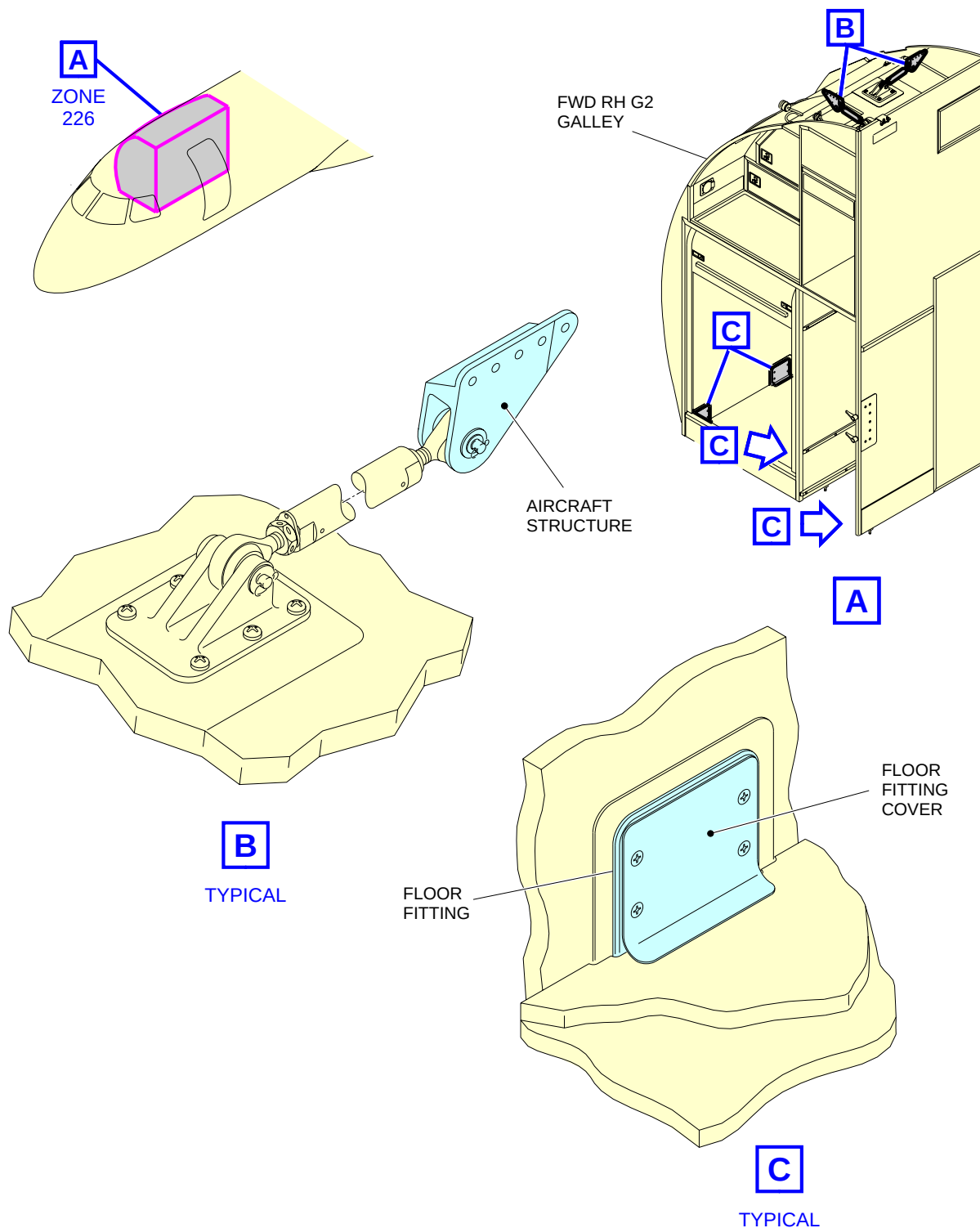
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EFFECTIVITY: ALL

Galley Attachments - Inspection

Figure 603 - Sheet 2



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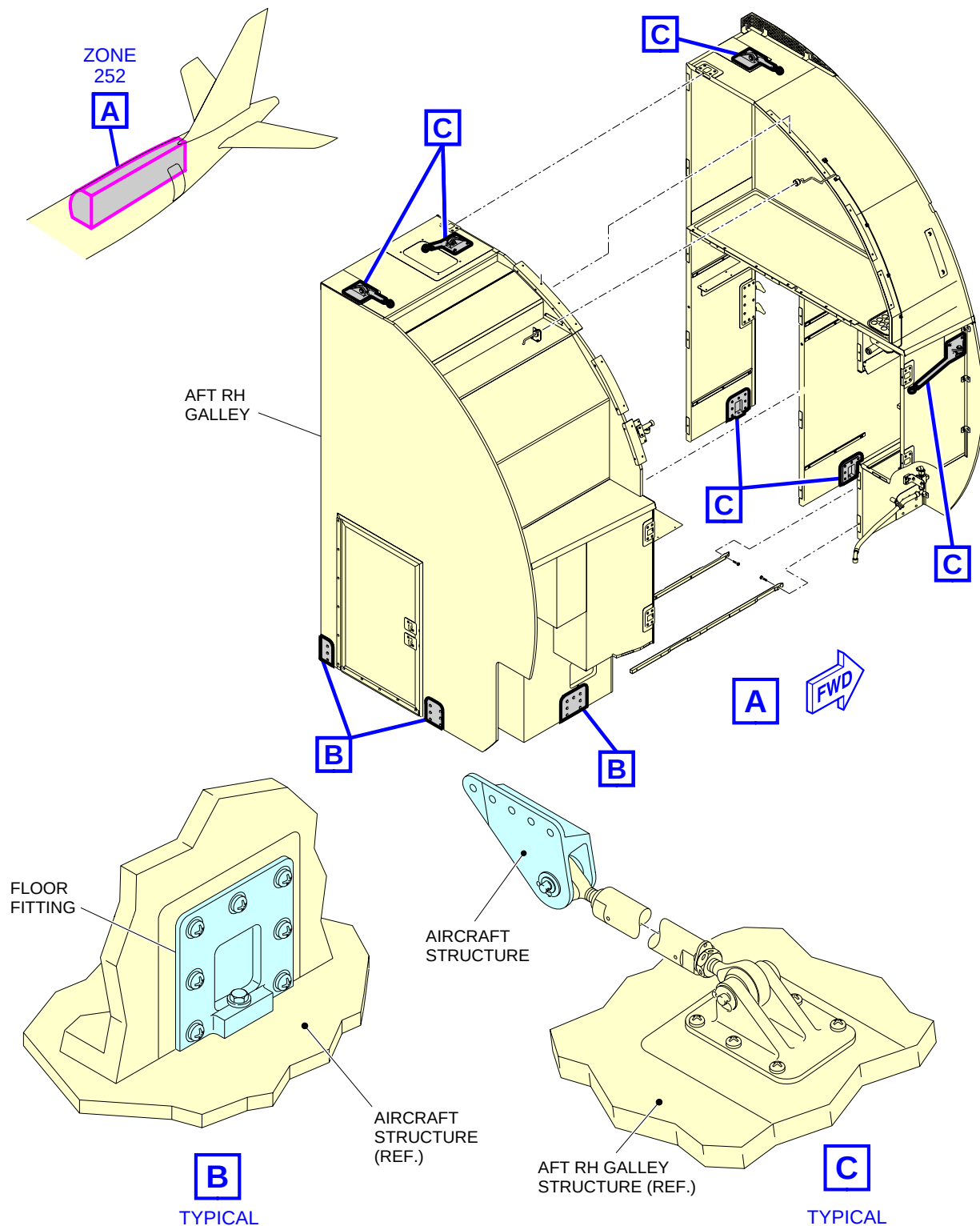
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Galley Attachments - Inspection

Figure 603 - Sheet 3



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